

# Challenge Write-Up Template

*Instructions: Copy this template for every challenge you submit. Fill in every section — a write-up that only shows the final flag without explanation will not receive full credit. Delete these italicized instruction lines before submitting.*

## Challenge Information

|                     |                            |
|---------------------|----------------------------|
| Student Name        | <i>Facon, Janna Mae L.</i> |
| Section / Class     | <i>BSIT- 3G</i>            |
| Challenge Name      | <i>Piece by Piece</i>      |
| Category            | <i>General Skills</i>      |
| Points / Difficulty | <i>Easy</i>                |
| Date Solved         | <i>8/13/2026</i>           |
| Flag Format Given   | <i>PicoCTF{...}</i>        |

## 1. Challenge Description

*Restate the challenge in your own words: what were you given, and what were you asked to find or do? Do not simply copy-paste the original prompt.*

The Piece by Piece challenge required me to connect to a remote Linux machine and recover a flag that had been divided into several separate files. The challenge provided multiple parts of a file that needed to be combined in the correct order to create a ZIP archive. After creating the archive, I had to extract it using the password provided in the instructions and then read the extracted flag.txt file to obtain the flag.

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## 2. Initial Analysis / Reconnaissance

*What did you check first? What did file, strings, a webpage view-source, or another first look tell you? What category did you decide this was, and why?*

The first step I did was read the challenge description to understand what I needed to accomplish. I learned that the flag was split into multiple parts and that these parts needed to be combined into a ZIP file. I then connected to the provided remote machine using SSH and used ls to inspect the files in my home directory. I found instructions.txt and the five parts of the file: part\_aa, part\_ab, part\_ac, part\_ad, and part\_ae. I read instructions.txt using cat instructions.txt, which explained that the parts needed to be combined into one file, that the resulting ZIP file was password protected, and that the password was **sunset**. This showed that the challenge involved basic Linux file manipulation, file concatenation, and extracting a password-protected ZIP archive.

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### 3. Tools and Commands Used

List every tool/command you used, in the order you used them, and what each one told you.

| Tool / Command                  | What it was used for                                                                             | Result / Output                                                                           |
|---------------------------------|--------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|
| ls                              | Checked the files available in the challenge directory.                                          | Found alice.txt, check_password.py, cupp.cfg, cupp.py, hash.txt, and userinfo.txt.        |
| less alice.txt                  | Examined the generated password list.                                                            | Displayed many possible password combinations based on the target's information.          |
| python check_password.py        | Ran the provided password-checking script.                                                       | Initially returned an error because passwords.txt was not found.                          |
| Password list generation        | Created the password list required by the checking script using the provided target information. | Generated candidate passwords for testing against the SHA-1 hash.                         |
| python check_password.py        | Tested the generated passwords against the target hash.                                          | Successfully found the password: picoCTF{Aj_15901990}.                                    |
| echo -n 'Aj_15901990'   sha1sum | Verified the SHA-1 hash of the recovered password.                                               | Produced 968c23490427d5d7debe72385a2c00eca5, matching the hash provided by the challenge. |

### 4. Solution Walkthrough

Explain your reasoning step by step — not just what you typed, but why you tried it and what you concluded from the result. Include screenshots, code snippets, or command output where useful. A reader who has not seen the challenge should be able to follow your logic and arrive at the same answer.

#### Step 1

The first step I did was read the challenge description. I learned that the flag had been split into multiple parts and that I needed to combine them and extract the final flag. I connected to the remote machine using SSH with the provided host and port. After logging in, I used `ls` and found five file parts named `part_aa`, `part_ab`, `part_ac`, `part_ad`, and `part_ae`, along with `instructions.txt`. I then used: `cat instructions.txt` The instructions explained that the flag was split into multiple parts, that the parts should be combined into one file, and that the resulting ZIP file was protected with the password **sunset**.

# Piece by Piece

General Skills Easy

by Yahaya Meddy · picoCTF 2026

After logging in, you will find multiple file parts in your home directory. These parts need to be combined and extracted to reveal the flag.

INSTANCE 1

Launch Instance

Additional details appear once you launch your instance.

FLAG 1

...

Submit

Created with CTFd by picoCTF 2026

Bookmark

9,868

97%

Open in Workspace

## Step 2

Since the files were named sequentially from part\_aa through part\_ae, I combined them in that order using the cat command: `cat part_aa part_ab part_ac part_ad part_ae > archive.zip` This redirected the combined contents of all five parts into a new file called archive.zip. I then checked the directory with ls and confirmed that archive.zip had been created. I also used: `file archive.zip` to verify that the combined file was recognized as a ZIP archive. This confirmed that the file parts had been combined correctly.

The screenshot shows the CTFd interface for the 'Piece by Piece' challenge. The challenge description states: 'After logging in, you will find multiple file parts in your home directory. These parts need to be combined and extracted to reveal the flag.' The 'Launch Instance' button is visible. The 'FLAG' field contains the text 'picoCTF{p1c\_e, a\_r\_c\_h\_i\_v\_e, \_z\_i\_p\_2786048b}'. To the right, a terminal window displays the following commands and output:

```
ls
part_aa  part_ab  part_ac  part_ad  part_ae
cat part_aa part_ab part_ac part_ad part_ae > archive.zip
ls
archive.zip
file archive.zip
archive.zip: Zip archive data, at least v1.0 to extract
```

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### Step 3

The instructions stated that the ZIP archive was password protected and that the password was **sunset**. I therefore extracted the archive using: `unzip -P sunset archive.zip` The extraction was successful and created `flag.txt`. I then used: `cat flag.txt` to read the contents of the extracted file. The terminal displayed:  
`picoCTF{z1p_and_spl1t_f1l3s_4r3_fun_27804340}` I copied this flag and entered it into the **FLAG** field on the Piece by Piece challenge page.



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## 5. Flag

`picoCTF{z1p_and_spl1t_f1l3s_4r3_fun_27804340}`

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## 6. Key Concept / What I Learned

*In 2-4 sentences, name the underlying security concept or vulnerability class this challenge taught (e.g. "weak RSA exponent," "reflected XSS," "stack buffer overflow"). This is the part of the write-up most worth remembering after the flag is submitted.*

*This challenge taught me about **file splitting, concatenation, and archive extraction in Linux**. I learned that separate file parts can be joined together in the correct order using `cat` and redirected into a new file. I also learned how to identify a ZIP archive and extract a password-protected archive using `unzip -P`.*

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## 7. References

*List any external resource you consulted beyond course materials (a tool's documentation, a technique explainer, a similar public write-up). It is normal and expected to use references — just cite them.*

[https://youtu.be/\\_Qqe9tmHHeI?si=zw12JZ-PHPjiBf27](https://youtu.be/_Qqe9tmHHeI?si=zw12JZ-PHPjiBf27)

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